Puncturing Linear Key Recovery Attacks

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Objective of this Presentation

- Understanding the motivation and core principles of puncturing
- **New** more accurate data complexity estimates
- Application and attack design examples based around real attacks
- Discussion of open problems

Contents of this Presentation

This talk summarizes the contributions of the following publications:

- Improving Linear Key Recovery Attacks using Walsh Spectrum Puncturing
  Antonio Flórez-Gutiérrez and Yosuke Todo, *EUROCRYPT 2024*
- Improving Linear Key Recovery Attacks using Walsh Spectrum Puncturing
  Antonio Flórez-Gutiérrez and Yosuke Todo, *Journal of Cryptology 38,4*
- Improved Cryptanalysis of ChaCha: Beating PNBs with Bit Puncturing
  Antonio Flórez-Gutiérrez and Yosuke Todo, *EUROCRYPT 2025*
- Divide-and-Conquer Trail Enumeration Puncturing: Application to Salsa and ChaCha
  Antonio Flórez-Gutiérrez and Yosuke Todo, *ASIACRYPT 2025*
- Accurate Parameter Estimates for Punctured Key Recovery Linear Attacks
  Tim Beyne, Antonio Flórez-Gutiérrez and Yosuke Todo, *ePrint 2026 (soon)*

Layout of this Presentation

1. Preliminaries and related work
2. Statistical model for puncturing
3. How to apply puncturing
4. Conclusion

Preliminaries

- Linear cryptanalysis basics
- (Pseudo)boolean functions and their Fourier transform
- Precedents for puncturing: plaintext Rejection and PNBs

Linear Approximations

Let $E : \mathbb{F}_2^n \times \mathbb{F}_2^\kappa \longrightarrow \mathbb{F}_2^n$ be a block cipher, $E(x, K) = E_K(x) = y$.

A linear approximation (Matsui, 1993) is a linear combination of bits of the plaintext and the ciphertext (and sometimes also the key) with masks α and β:

$$\langle \alpha, x \rangle \oplus \langle \beta, y \rangle$$

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$$\langle \alpha, x \rangle \oplus \langle \beta, y \rangle$$

The correlation measures the statistical imbalance of the approximation:

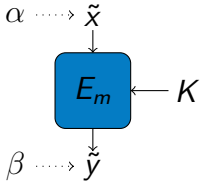
$$\mathrm{cor}_K(\alpha, \beta) = \frac{1}{2^n} \sum_{x \in \mathbb{F}_2^n} (-1)^{\langle \alpha, x \rangle \oplus \langle \beta, E_K(x) \rangle}$$

Note that the correlation is, in principle, key-dependent, but we assume it's c.

We can distinguish E with a sample of $N \approx 1/c^2$ plaintext-ciphertext pairs.

Linear Key Recovery Attack

Partial approximation: $\langle \alpha, \tilde{x} \rangle \oplus \langle \beta, \tilde{y} \rangle$ with correlation c



Linear Key Recovery Attack

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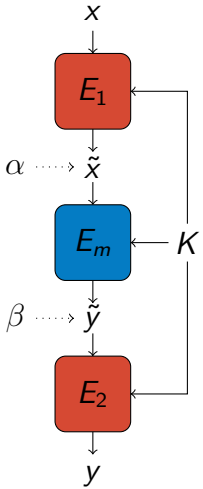
We express the linear approximation as a function of the plaintext, ciphertext and key with the key recovery map, for example:

$$E_1^{\text{trunc}}(x, k) \oplus E_2^{\text{trunc}}(y, k), \text{ where } k \text{ is part of the key}$$

This function can be abstracted to a map of the form $f(x, k)$



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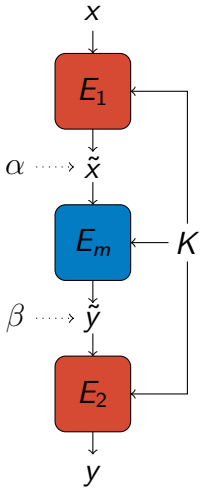
We compute the experimental correlations for all guesses of k:

$$\widetilde{\text{cor}}(k) = \frac{1}{N} \sum_{x \in \mathcal{D}} (-1)^{f(x,k)}, \ \mathcal{D} \text{ of size } N \approx 1/c^2$$

The correct guess is expected to exhibit a larger correlation.



Linear Key Recovery Attack *(cont.)*

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  Uses a distillation phase to extract the important information from the data in the beginning and avoid redundant computations.
- $N + |k_O| \cdot 2^{|k_I|+|k_O|}$ (Collard et al., 2007)
  Leverages the Fast Walsh Transform algorithm for optimization.

(Pseudo)boolean functions

Let $\mathbb{F}_2 = \{0, 1\}$ denote the binary field and $\mathbb{R}$ denote the real numbers.

Definition: Pseudoboolean function

Pseudoboolean functions are maps of the form $f : \mathbb{F}_2^\ell \longrightarrow \mathbb{R}$ (alternatively $\mathbb{C}$). They form a vector space denoted $\mathbb{R}\mathbb{F}_2^\ell$ which can be identified with $\mathbb{R}^{2^\ell}$.

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$\mathbb{R}\mathbb{F}_2^\ell$ has an inner product and the associated 2-norm:

$$\langle f, g \rangle = \frac{1}{2^\ell} \sum_{x \in \mathbb{F}_2^\ell} f(x)g(x), \qquad \|f\|_2 = \sqrt{\langle f, f \rangle}.$$

We also say that f and g are orthogonal if $\langle f, g \rangle = 0$ and write $f \perp g$.

Fourier Transform

Definition: Hadamard basis

The **Hadamard basis** of $\mathbb{RF}_2^\ell$ consists of the parity functions $\mathbf{h}_u(x) = (-1)^{\langle u,x\rangle}$.

This is an orthonormal basis ($\langle \mathbf{h}_u, \mathbf{h}_v \rangle = 1$ if $u = v$ and 0 otherwise).

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Definition: Fourier transform

Given $f : \mathbb{F}_2^\ell \longrightarrow \mathbb{R}$, its **Fourier transform** is the map $\widehat{f} : \mathbb{F}_2^\ell \longrightarrow \mathbb{R}$ given by

$$\widehat{f}(u) = \frac{1}{2^\ell} \sum_{x \in \mathbb{F}_2^\ell} (-1)^{\langle u,x\rangle} f(x)$$

This represents a function in the Hadamard basis, $f = \sum_{u \in \mathbb{F}_2^\ell} \widehat{f}(u) \mathbf{h}_u$.

Fourier Transform *(cont.)*

The Fourier transform codifies a lot of information about f, for example:

- $\langle f, g \rangle = 2^{\ell} \langle \widehat{f}, \widehat{g} \rangle$ and $\|f\|_2 = \sqrt{2^{\ell}} \|\widehat{f}\|_2$.
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- If f is a Boolean function, $\widehat{f}(0)$ is the correlation of f.
- Furthermore, $\widehat{f}(0)$ is average $\mathrm{E}[f(x)]$ of the variable $f(x)$ for random x.
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- $\mathrm{Var}(f(x)) = \|f\|_2^2 - \widehat{f}(0)^2$
- The value of the function f is not influenced by the input bit in position i if and only if $\widehat{f}(u) = 0$ for all u with $u_i = 1$.

Correlation Matrices and Linear Trails

Definition: Correlation matrices

Let $f : \mathbb{F}_2^n \longrightarrow \mathbb{F}_2^m$ be a vectorial Boolean function. Its **correlation matrix** is

$$\widehat{f}(u, v) = \widehat{f_v}(u) = \frac{1}{2^n} \sum_{x \in \mathbb{F}_2^n} (-1)^{\langle u, x \rangle \oplus \langle v, f(x) \rangle}$$

If $f = f_r \circ \cdots \circ f_1$, then $\widehat{f} = \widehat{f_r} \times \cdots \times \widehat{f_1}$.

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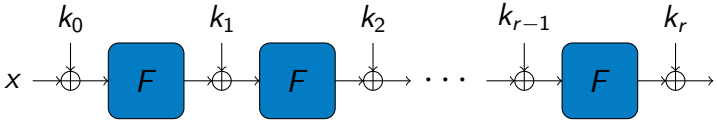
For a block chipher E, we have $\mathrm{cor}_K(\alpha, \beta) = \widehat{E_K}(\alpha, \beta)$.
This means that for iterative block ciphers $E_K = E_r^{K_r} \circ \cdots \circ E_1^{K_1}$:

$$\mathrm{cor}_K(\alpha_r, \alpha_0) = \sum_{\alpha_1} \cdots \sum_{\alpha_{r-1}} \prod_{i=1}^{r} \widehat{E_i^{K_i}}(\alpha_{i-1}, \alpha_i)$$

$(\alpha_0, \alpha_1, \ldots, \alpha_r)$ is called a linear trail.

Correlation Matrices and Linear Trails *(cont.)*

Consider the case of key-alternating block ciphers.



Then the key only flips the sign of a trail's correlation contribution:

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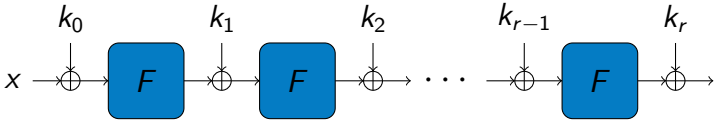

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If there are many trails of similar correlation, we have $\mathrm{E}\left[\mathrm{cor}_K(\alpha_0, \alpha_r)\right] = 0$ and

$$\mathrm{Var}\left(\mathrm{cor}_K(\alpha_0, \alpha_r)\right) = \mathrm{ELP}(\alpha_0, \alpha_r) = \sum_{\alpha_1} \cdots \sum_{\alpha_{r-1}} \prod_{i=1}^{r} \widehat{F}(\alpha_{i-1}, \alpha_i)^2$$

Statistics of Linear Cryptanalysis

The data complexity is determined by the different distributions of $\widetilde{\mathrm{cor}}$ under right and wrong key. In the model of (Blondeau and Nyberg, 2017), they depend on the random correct key K and the random data sample of N plaintexts.

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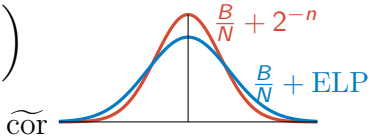
For the experimental correlation with N data pairs:

$$\widetilde{\mathrm{cor}}_R \sim_{K,\mathcal{D}} \mathcal{N}\left(0, \frac{B}{N} + \mathrm{ELP}\right) \quad \widetilde{\mathrm{cor}}_W \sim_{K,\mathcal{D}} \mathcal{N}\left(0, \frac{B}{N} + 2^{-n}\right)$$

$B = \frac{2^n - N}{2^n - 1}$ for distinct data pairs and $B = 1$ otherwise.



Plaintext Rejection

When designing a linear attack, we often want to replace the key recovery map f for an "easier" function g, for example by reducing the number of active key bits.

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Intuition: If we keep a fraction ρ of the data, we increase the sample size by $1/\rho$.

Reality: While this is true for small data complexities, it becomes quite a bit more complicated when approaching the full codebook (Wu et al., 2024).

Prohabilistic Neutral Bits (PNBs)

(Aumasson et al., 2008), differential-linear attacks on Salsa/ChaCha

Idea: Because of the ARX construction, there are many active bits in the final key addition, but some are much more significant than others.

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Idea: Because of the ARX construction, there are many active bits in the final key addition, but some are much more significant than others.

1. Considering each key bit k^i, flip this bit and check the correlation between $f(x, k)$ and $f(x, x \oplus k^i)$ experimentally for a large data/key sample.
2. If this correlation is above a threshold, the bit is marked as a PNB, otherwise it is marked as a significant bit.
3. Perform a second experiment to determine the correlation between $f(x, k)$ and a reduced version which sets the PNBs to random fixed values.

Motivation for Puncturing

In plaintext rejection, we replace f for $g : \mathbb{F}_2^\ell \longrightarrow \{1, -1, 0\}$

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What if we replace f for an arbitrary pseudoboolean function g?

Statistical Model

- Approximation data complexity result
- Correct approximation data complexity result
- The case of puncturing

Approximation Result

Theorem: Key Recovery Map Approximation

Consider that the key recovery map $f : \mathbb{F}_2^\ell \longrightarrow \mathbb{R}$ is substituted by the approximation $g : \mathbb{F}_2^\ell \longrightarrow \mathbb{R}$. The **Pearson correlation coefficient** ρ is

$$\rho = \frac{\langle f, g \rangle}{\|f\|_2 \cdot \|g\|_2} = \frac{\langle \widehat{f}, \widehat{g} \rangle}{\|\widehat{f}\|_2 \cdot \|\widehat{g}\|_2}$$

Then, the distributions of the experimental correlations using g are:

$$\widetilde{\mathrm{cor}}_R^g \sim \|g\|_2 \cdot \mathcal{N}\left(0, \frac{B}{N} + \rho^2 \cdot \mathrm{ELP}\right)$$

$$\widetilde{\mathrm{cor}}_W^g \sim \|g\|_2 \cdot \mathcal{N}\left(0, \frac{B}{N} + \rho^2 \cdot 2^{-n}\right)$$

Approximation Result (new version)

Theorem: Key Recovery Map Approximation **(new)**

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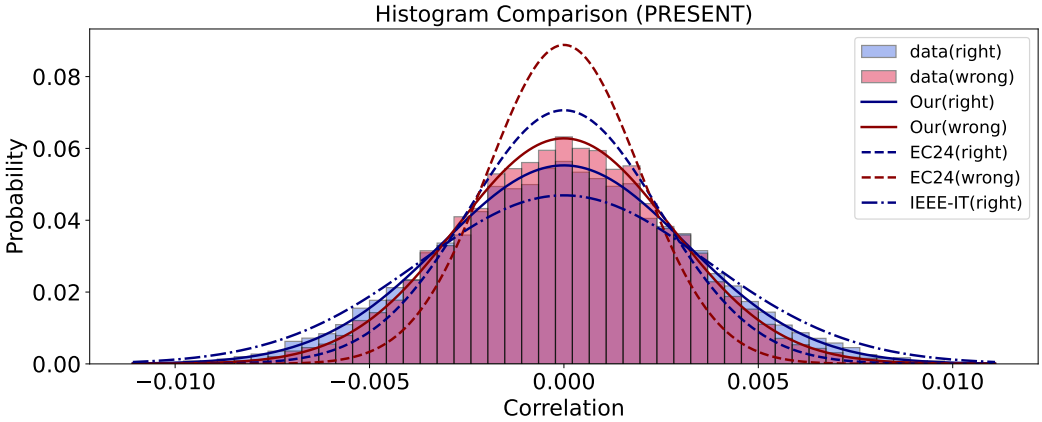
Then, the distributions of the experimental correlations using g are:

$$\widetilde{\mathrm{cor}}_R^g \sim \|g\|_2 \cdot \mathcal{N}\left(0, \frac{B}{N} + \rho^2 \cdot \mathrm{ELP} + (1-\rho^2)2^{-n}\right)$$

$$\widetilde{\mathrm{cor}}_W^g \sim \|g\|_2 \cdot \mathcal{N}\left(0, \frac{B}{N} + 2^{-n}\right)$$

Experimental results



Experiment on 16-bit toy present using the full codebook

Data complexity compensation

In (Flórez-Gutiérrez and Todo, 2024), the following simplified rule is proposed:

Corollary: Data complexity **(old)**

The success probability of a linear attack which substitutes the key recovery map f for the balanced approximation g remains the same as long as the corrected data sample size N/B is increased by a factor $1/\rho^2$.

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Here, we suggest the following alternative:

Corollary: Data complexity **(new)**

In a linear attack which substitutes the key recovery map f for the balanced approximation g, and assuming the linear approximation has average correlation $c = 0$, the data complexity, success probability and advantage of the attack are equivalent to substituting the ELP for $\rho^2 \cdot \mathrm{ELP} + (1 - \rho^2)2^{-n}$.

Puncturing

Definition: Key Recovery Map Walsh Spectrum Puncturing

Given the key recovery map $f : \mathbb{F}_2^\ell \longrightarrow \mathbb{F}_2$, a **puncture set** is any $\mathcal{P} \subseteq \mathbb{F}_2^\ell$. We define $g : \mathbb{F}_2^\ell \longrightarrow \mathbb{R}$ as a function whose Fourier transform is:

$$\widehat{g}(u) = \begin{cases} \widehat{f}(u) & \text{if } u \notin \mathcal{P} \\ 0 & \text{if } u \in \mathcal{P} \end{cases}$$

The "proportion" of f which has been removed/kept is determined by:

$$\epsilon = \sum_{u \in \mathcal{P}} \widehat{f}(u)^2, \qquad \eta = \sum_{u \notin \mathcal{P}} \widehat{f}(u)^2, \qquad \epsilon + \eta = 1$$

ϵ is called **puncturing coefficient** while η is called **puncturing correlation**.

Puncturing *(cont.)*

Noting that $\eta = 1 - \epsilon = \rho^2$, we deduce that:

Theorem: Puncturing Data Complexity **(old)**

In puncturing, the corrected data complexity is increased by a factor of $\frac{1}{1-\epsilon} = \frac{1}{\eta}$.

This result is accurate in most settings, but when the data complexity is close to the full codebook, the following version should be used instead:

Theorem: Puncturing Data Complexity **(new)**

In puncturing, the data complexity, advantage and success probability can be estimated by assuming that the approximation's ELP is $\eta \cdot \mathrm{ELP} + (1 - \eta)2^{-n}$.

Puncturing can be shown to be optimal.

Applications

- Overview of puncturing strategies
- Bit puncturing: application to GIFT
- Trail enumeration puncturing: application to ChaCha

Overview of puncturing strategies

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The main challenge when designing attacks with puncturing is deciding which positions to set to zero: the input domain of f is too large to build $\widehat{f}$.

- For Sbox-based ciphers, we can restrict the Sbox Fourier transform to an affine subspace or remove coefficients of large Hamming weight.
- Similarly to PNBs, we can remove certain bits from the input of f or its components. This is called bit puncturing.
- We can work on an approximation of $\widehat{f}$ built by accumulating significant linear trails of f. This is trail enumeration puncturing.

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PNBs

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In order to puncture bit i, we follow these steps:

PNBs

1. Take $\widehat{f}$ and compute $\widehat{g}$ as:

$$\widehat{g}(u) = \widehat{f}(u) \text{ if } u_i = 0, \quad \widehat{g}(u) = 0 \text{ if } u_i = 1$$

Bit Puncturing

2. g is f's average over both values of bit i:

$$g(x) = \frac{1}{2}\left(f(x) + f(x \oplus \mathrm{e}_i)\right)$$

3. $\eta = \sum_{u_i=0} \widehat{f}(u)^2$

Example: GIFT-128

128-bit PRESENT-like cipher, (Banik et al., 2017)

The attack uses a linear approximation found by (Wang et al., 2022):

$$0000\ 0000\ 0000\ 0000\ 0202\ 0000\ 0000\ 0000$$

$$\longrightarrow \text{ 17 rounds, } c \sim \mathcal{N}(\pm 2^{-55.42}, 2^{-110.57}) \longrightarrow$$

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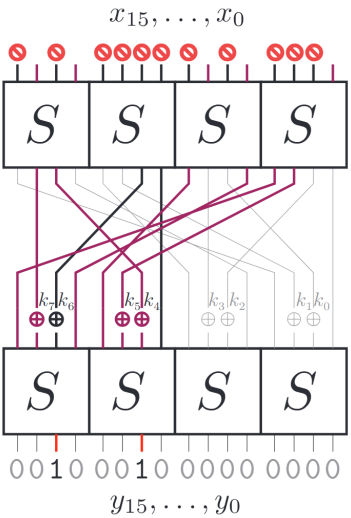
$$\longrightarrow \text{ 17 rounds, } c \sim \mathcal{N}(\pm 2^{-55.42}, 2^{-110.57}) \longrightarrow$$

$$0044\ 0000\ 0022\ 0000\ 0000\ 0000\ 0000\ 0000$$

The general structure of the attack is:

3R KR $\rightarrow$ 2R bit-punctured KR $\rightarrow$ 17R approx. $\rightarrow$ 3R KR

Example: GIFT-128 *(cont.)*

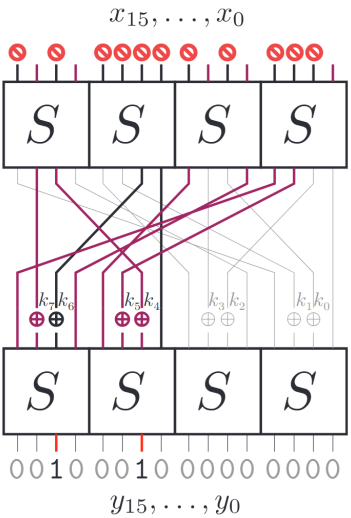


Target: output bit $y_{13} \oplus y_9$ of the super Sbox

Puncture input bits $x_{15}, x_{12}, x_{10} \ldots x_7, x_5, x_3 \ldots x_1$

In theory, there are 8 possible punctured spectra depending on (k_4, k_5, k_7), but in reality it is determined by $k_5 \oplus k_4$, and we always have $\eta = 2^{-8.30}$.

Example: GIFT-128 *(cont.)*



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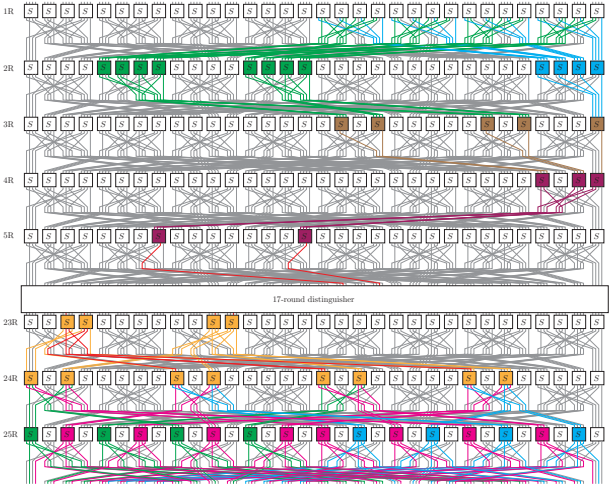
In summary, we have gone from a function with 16 input bits and a 4-bit internal key guess to 5 input bits and a 1-bit internal key guess, at the cost of roughly $2^{8.30}$ times the data.

Example: GIFT-128 *(cont.)*



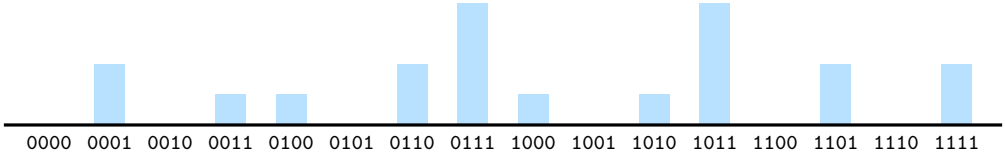
By applying bit puncturing to the super Sbox in rounds 4 and 5, the diffusion of the key recovery map in rounds 1 through 3 is greatly reduced.

The data complexity is 2^{123}.

Using a partial sum strategy for the key recovery, the overall time complexity is $2^{124.6}$.

Trail Enumeration Puncturing

Sometimes the key recovery map has an input domain which is too large and can't be divided easily into components which can be punctured.



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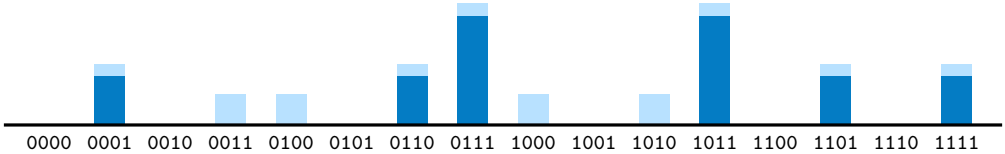
However, we can use a linear trail enumeration algorithm on f and approximate each coefficient in $\widehat{f}$ as the sum of the correlations of its dominant trails.

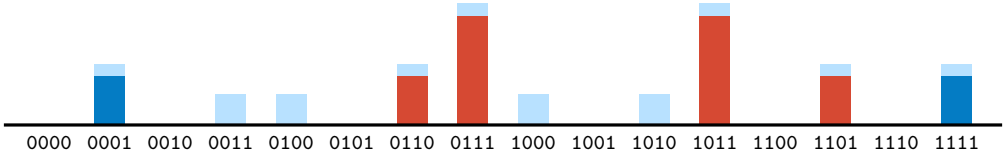


Trail Enumeration Puncturing

Sometimes the key recovery map has an input domain which is too large and can't be divided easily into components which can be punctured.

However, we can use a linear trail enumeration algorithm on f and approximate each coefficient in $\widehat{f}$ as the sum of the correlations of its dominant trails.

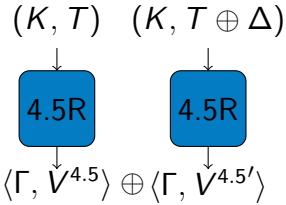
Then we can puncture this approximation to $\widehat{g}$ and take $\eta = \sum \widehat{g}(u)^2$.



Example: ChaCha

(Bernstein,2008), stream cipher based on 512-bit ARX permutation, most attacks follow (Aumasson et al., 2008)

Differential-linear distinguisher: Under input difference Δ, the mask Γ exhibits a correlation of $2^{-13.3}$ after 4.5 rounds.



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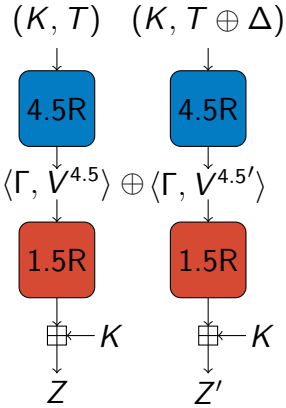
Differential-linear distinguisher: Under input difference Δ, the mask Γ exhibits a correlation of $2^{-13.3}$ after 4.5 rounds.

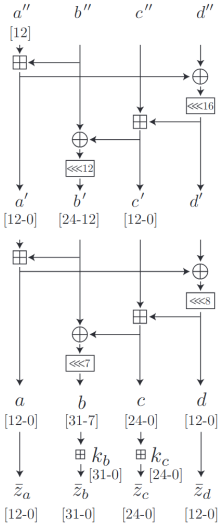
1.5 rounds of key recovery are appended for 6 total rounds.

However, the diffusion at each modular addition is very broad, which means most of the key is active.

Attacks solve this with PNBs, which are not optimal.

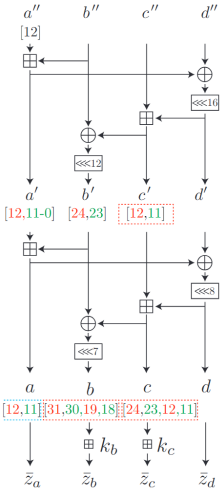


Example: ChaCha *(cont.)*



Consider a simplified example: even after just one round, one bit activates 57 key bits with normal propagation rules.

Example: ChaCha *(cont.)*



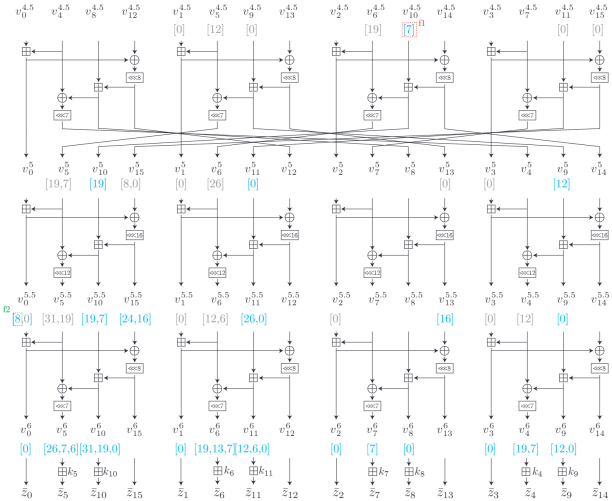
Consider a simplified example: even after just one round, one bit activates 57 key bits with normal propagation rules.

We control the propagation round by round:

- After the first half, we keep 26 trails: red bits are always active, and green bits are active in some but not all trails, we have $\eta = 2^{-0.58}$.
- After the second half: 8 masks, $\eta = 2^{-1.17}$
- After the key addition: 768 masks, $\eta = 2^{-6.17}$

Finally, we have 8 active key bits, and only 4 are nonlinear.

ChaCha *(cont.)*



The real key recovery is a little more complicated...

We propagate the mask as far as possible and write it as the XOR of 20 functions we assume independent (blue bits).

The two most complicated functions f_1 and f_2 are punctured.

In the end, the attack uses data $2^{51.0}$ and time $2^{61.4}$.

Divide-and-conquer Trail Enumeration

When applying trail enumeration to Salsa, the number of trails grows too fast.

In the ChaCha attack, we split f as a XOR of functions, what if we went further?

Divide-and-conquer Trail Enumeration

When applying trail enumeration to Salsa, the number of trails grows too fast.

In the ChaCha attack, we split f as a XOR of functions, what if we went further?

Domain Separation Decomposition

$$f = \bigoplus f_i = \prod f_i$$

$$\eta = \|g\|_2^2 = \prod \|g_i\|_2^2$$

Spectrum Linear Decomposition

$$f = \sum f_\lambda$$

$$\eta = \|g\|_2^2 = \sum \sum \langle g_\lambda, g_{\lambda'} \rangle$$

We alternate between both decompositions until trail enumeration is possible.

Conclusion

- Summary of published attacks using puncturing
- Main takeaways
- Where to go next?

Summary of Attacks

Target	Rounds	Data	Time	Memory	P_S
Serpent (192-bit)	12	$2^{127.50}$ DKP	$2^{189.74}$	$2^{182.00}$	80%
Serpent (256-bit)	12	$2^{125.16}$ DKP	$2^{214.36}$	$2^{125.16}$	85%
	12	$2^{126.30}$ DKP	$2^{210.36}$	$2^{125.16}$	85%
GIFT-128 (General)	25	$2^{123.02}$ KP	$2^{124.61}$	$2^{112.00}$	79%
GIFT-128 (COFB)	7	$2^{62.10}$ KP	$2^{125.09}$	$2^{62.10}$	80%
GIFT-128 (HyENA)	7	$2^{62.10}$ KP	$2^{125.38}$	$2^{62.10}$	80%
DES	Full	$2^{41.62}$ KP	$2^{41.76}$	$2^{34.54}$	62%
NOEKEON	12	$2^{119.55}$ KP	$2^{120.63}$	$2^{115.00}$	79%

Summary of Attacks *(cont.)*

Target	Rounds	Data	Time	P_S
Salsa128	7	$2^{54.8}$ CIV	$2^{86.7}$	99%
	8	$2^{110.3}$ CIV	$2^{125.8}$	93%
Salsa256	8.5	$2^{126.7}$ CIV	$2^{219.7}$	80%
ChaCha256	6	$2^{51.0}$ CIV	$2^{61.4}$	99%
		$2^{55.7}$ CIV	$2^{57.4}$	99%
	7	$2^{127.7}$ CIV	$2^{148.2}$	99%
		$2^{102.9}$ CIV	$2^{154.2}$	99%
	7.5	$2^{127.1}$ CIV	$2^{243.6}$	82%

Summary and Observations

- Puncturing enables us to design efficient key recovery attacks by ignoring key guess bits which are not really significant to the attack.
- In particular, it provides a powerful alternative to PNBs.
- The first model for puncturing is inaccurate for very high data complexities, but can still be used as a rough estimate when designing attacks.
- The core challenge when designing attacks is choosing where to puncture: we have several techniques but they take a long time to implement.

Future Research Directions

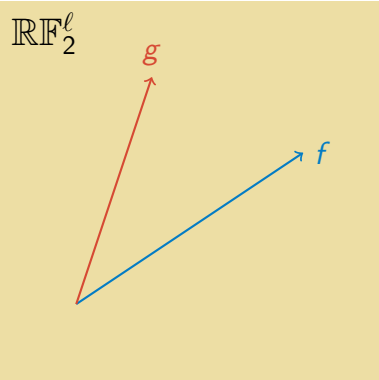
- Further study of puncturing strategies, as well as developing tools for designing them, would greatly simplify future applications.
- The current model can only leverage one linear approximation: using more than one might not just increase the accuracy but lead to better attacks.
- The issue of correct and optimal application to multiple and multidimensional attacks also remains open.
- Puncturing blurs the line between the distinguisher and the key recovery parts of an attack: is it possible to mix them completely?

Appendix

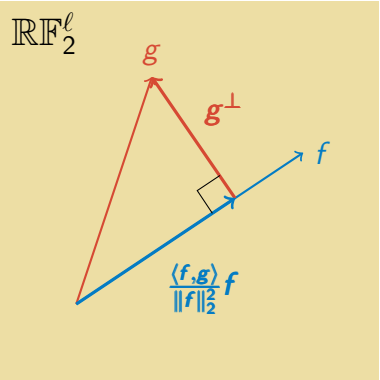
- Old proof of the pproximation data complexity result
- New proof of the approximation data complexity result

Sketch of the Proof

We think of f and g as vectors in $\mathbb{R}^{2^\ell}$.



Sketch of the Proof



We think of f and g as vectors in $\mathbb{R}^{2^\ell}$.
We can project g onto f orthogonally:

$$g = \frac{\langle f, g\rangle}{\|f\|_2^2} f + g^\perp \text{ with } g^\perp \perp f$$

From Pythagoras' theorem, we know that

$$\|g\|_2^2 = \frac{\langle f, g\rangle^2}{\|f\|_2^2} + \|g^\perp\|_2^2$$

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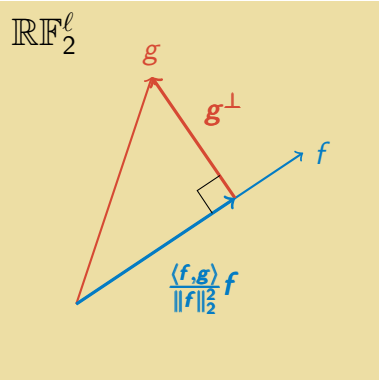

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This extends to the experimental correlation:

$$\widetilde{\mathrm{cor}}^g = \frac{\langle f, g\rangle}{\|f\|_2^2} \widetilde{\mathrm{cor}}^f + \widetilde{\mathrm{cor}}^{g^\perp}$$

Sketch of the Proof *(cont.)*

We will first compute the distribution for the right key statistic:

$$\mathrm{E}_{\mathcal{D},K}\left[\widetilde{\mathrm{cor}}_R^g\right] =$$

$$\mathrm{Var}_{\mathcal{D},K}\left(\widetilde{\mathrm{cor}}_R^g\right) =$$

Sketch of the Proof *(cont.)*

We will first compute the distribution for the right key statistic:

$$\mathrm{E}_{\mathcal{D},K}\left[\widetilde{\mathrm{cor}}_R^g\right] = \mathrm{E}_{\mathcal{D},K}\left[\frac{\langle f, g\rangle}{\|f\|_2^2}\widetilde{\mathrm{cor}}_R^f\right] + \mathrm{E}_{\mathcal{D},K}\left[\widetilde{\mathrm{cor}}_R^{g^\perp}\right]$$

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Sketch of the Proof *(cont.)*

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Sketch of the Proof *(cont.)*

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$$+ \mathrm{E}_{\mathcal{K}}\left[\mathrm{Var}_{\mathcal{D}}\left(\widetilde{\mathrm{cor}}_R^{g^\perp}\right)\right] - 2\mathrm{Cov}_{\mathcal{D},\mathcal{K}}\left(\frac{\langle f, g\rangle}{\|f\|_2^2}\widetilde{\mathrm{cor}}_R^f, \widetilde{\mathrm{cor}}_R^{g^\perp}\right)$$

Sketch of the Proof *(cont.)*

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$$\mathrm{Var}_{\mathcal{D},K}\left(\widetilde{\mathrm{cor}}_R^g\right) = \mathrm{Var}_{\mathcal{D},K}\left(\frac{\langle f, g\rangle}{\|f\|_2^2}\widetilde{\mathrm{cor}}_R^f\right) + \cancel{\mathrm{Var}_K\left(\mathrm{E}_{\mathcal{D}}\left[\widetilde{\mathrm{cor}}_R^{g\perp}\right]\right)}$$

$$+ \mathrm{E}_K\left[\mathrm{Var}_{\mathcal{D}}\left(\widetilde{\mathrm{cor}}_R^{g\perp}\right)\right] - \cancel{2\mathrm{Cov}_{\mathcal{D},K}\left(\frac{\langle f, g\rangle}{\|f\|_2^2}\widetilde{\mathrm{cor}}_R^f, \widetilde{\mathrm{cor}}_R^{g\perp}\right)}$$

Sketch of the Proof *(cont.)*

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Sketch of the Proof *(cont.)*

We will first compute the distribution for the right key statistic:

$$\mathrm{E}_{\mathcal{D},\mathcal{K}}\left[\widetilde{\mathrm{cor}}_R^g\right] = \cancel{\mathrm{E}_{\mathcal{D},\mathcal{K}}\left[\frac{\langle f, g\rangle}{\|f\|_2^2}\widetilde{\mathrm{cor}}_R^f\right]} + \cancel{\mathrm{E}_{\mathcal{D},\mathcal{K}}\left[\widetilde{\mathrm{cor}}_R^{g^\perp}\right]} \simeq 0$$

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&\simeq \frac{\langle f, g\rangle^2}{\|f\|_2^2}\left(\frac{B}{N} + \mathrm{ELP}\right) + \|g^\perp\|_2^2\frac{B}{N} \\
&= \|g\|_2^2\frac{B}{N} + \frac{\langle f, g\rangle^2}{\|f\|_2^2}\cdot\mathrm{ELP}
\end{aligned}$$

Sketch of the Proof *(cont.)*

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$$\mathrm{E}_{\mathcal{D},\mathcal{K}}\left[\widetilde{\mathrm{cor}}_R^g\right] = \cancel{\mathrm{E}_{\mathcal{D},\mathcal{K}}\left[\frac{\langle f, g\rangle}{\|f\|_2^2}\widetilde{\mathrm{cor}}_R^f\right]} + \cancel{\mathrm{E}_{\mathcal{D},\mathcal{K}}\left[\widetilde{\mathrm{cor}}_R^{g^\perp}\right]} \simeq 0$$

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&\simeq \frac{\langle f, g\rangle^2}{\|f\|_2^2}\left(\frac{B}{N} + \mathrm{ELP}\right) + \|g^\perp\|_2^2\frac{B}{N} \\
&= \|g\|_2^2\frac{B}{N} + \frac{\langle f, g\rangle^2}{\|f\|_2^2}\cdot\mathrm{ELP} = \|g\|_2^2\left(\frac{B}{N} + \rho^2\cdot\mathrm{ELP}\right)
\end{aligned}$$

Sketch of the Proof *(cont.)*

The wrong key case is handled in the same way:

$$\mathrm{E}_{\mathcal{D},\mathcal{K}}\left[\widetilde{\mathrm{cor}}_{W}^{g}\right] =$$

$$\mathrm{Var}_{\mathcal{D},\mathcal{K}}\left(\widetilde{\mathrm{cor}}_{W}^{g}\right) =$$

Sketch of the Proof *(cont.)*

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Sketch of the Proof *(cont.)*

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Sketch of the Proof *(cont.)*

The wrong key case is handled in the same way:

$$\mathrm{E}_{\mathcal{D},K}\left[\widetilde{\mathrm{cor}}_W^g\right] = \cancel{\mathrm{E}_{\mathcal{D},K}\left[\frac{\langle f, g\rangle}{\|f\|_2^2}\widetilde{\mathrm{cor}}_W^f\right]} + \cancel{\mathrm{E}_{\mathcal{D},K}\left[\widetilde{\mathrm{cor}}_W^{g^\perp}\right]} \simeq 0$$

$$\begin{aligned}\mathrm{Var}_{\mathcal{D},K}\left(\widetilde{\mathrm{cor}}_W^g\right) &= \mathrm{Var}_{\mathcal{D},K}\left(\frac{\langle f, g\rangle}{\|f\|_2^2}\widetilde{\mathrm{cor}}_W^f\right) + \cancel{\mathrm{Var}_K\left(\mathrm{E}_{\mathcal{D}}\left[\widetilde{\mathrm{cor}}_W^{g^\perp}\right]\right)} \\ &\quad + \mathrm{E}_K\left[\mathrm{Var}_{\mathcal{D}}\left(\widetilde{\mathrm{cor}}_W^{g^\perp}\right)\right] - \cancel{2\mathrm{Cov}_{\mathcal{D},K}\left(\frac{\langle f, g\rangle}{\|f\|_2^2}\widetilde{\mathrm{cor}}_W^f, \widetilde{\mathrm{cor}}_W^{g^\perp}\right)} \\ &\simeq \frac{\langle f, g\rangle^2}{\|f\|_2^2}\left(\frac{B}{N} + 2^{-n}\right) + \|g^\perp\|_2^2\frac{B}{N}\end{aligned}$$

Sketch of the Proof *(cont.)*

The wrong key case is handled in the same way:

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$$\begin{aligned}
\mathrm{Var}_{\mathcal{D},K}\left(\widetilde{\mathrm{cor}}_W^g\right) &= \mathrm{Var}_{\mathcal{D},K}\left(\frac{\langle f, g\rangle}{\|f\|_2^2}\widetilde{\mathrm{cor}}_W^f\right) + \cancel{\mathrm{Var}_K\left(\mathrm{E}_{\mathcal{D}}\left[\widetilde{\mathrm{cor}}_W^{g^\perp}\right]\right)} \\
&\quad + \mathrm{E}_K\left[\mathrm{Var}_{\mathcal{D}}\left(\widetilde{\mathrm{cor}}_W^{g^\perp}\right)\right] - \cancel{2\mathrm{Cov}_{\mathcal{D},K}\left(\frac{\langle f, g\rangle}{\|f\|_2^2}\widetilde{\mathrm{cor}}_W^f, \widetilde{\mathrm{cor}}_W^{g^\perp}\right)} \\
&\simeq \frac{\langle f, g\rangle^2}{\|f\|_2^2}\left(\frac{B}{N} + 2^{-n}\right) + \|g^\perp\|_2^2\frac{B}{N} \\
&= \|g\|_2^2\frac{B}{N} + \frac{\langle f, g\rangle^2}{\|f\|_2^2}\cdot 2^{-n}
\end{aligned}$$

Sketch of the Proof *(cont.)*

The wrong key case is handled in the same way:

$$\mathrm{E}_{\mathcal{D},K}\left[\widetilde{\mathrm{cor}}_W^g\right] = \cancel{\mathrm{E}_{\mathcal{D},K}\left[\frac{\langle f, g\rangle}{\|f\|_2^2}\widetilde{\mathrm{cor}}_W^f\right]} + \cancel{\mathrm{E}_{\mathcal{D},K}\left[\widetilde{\mathrm{cor}}_W^{g^\perp}\right]} \simeq 0$$

$$\begin{aligned}
\mathrm{Var}_{\mathcal{D},K}\left(\widetilde{\mathrm{cor}}_W^g\right) &= \mathrm{Var}_{\mathcal{D},K}\left(\frac{\langle f, g\rangle}{\|f\|_2^2}\widetilde{\mathrm{cor}}_W^f\right) + \cancel{\mathrm{Var}_K\left(\mathrm{E}_{\mathcal{D}}\left[\widetilde{\mathrm{cor}}_W^{g^\perp}\right]\right)} \\
&\quad + \mathrm{E}_K\left[\mathrm{Var}_{\mathcal{D}}\left(\widetilde{\mathrm{cor}}_W^{g^\perp}\right)\right] - \cancel{2\mathrm{Cov}_{\mathcal{D},K}\left(\frac{\langle f, g\rangle}{\|f\|_2^2}\widetilde{\mathrm{cor}}_W^f, \widetilde{\mathrm{cor}}_W^{g^\perp}\right)} \\
&\simeq \frac{\langle f, g\rangle^2}{\|f\|_2^2}\left(\frac{B}{N} + 2^{-n}\right) + \|g^\perp\|_2^2\frac{B}{N} \\
&= \|g\|_2^2\frac{B}{N} + \frac{\langle f, g\rangle^2}{\|f\|_2^2}\cdot 2^{-n} = \|g\|_2^2\left(\frac{B}{N} + \rho^2\cdot 2^{-n}\right)
\end{aligned}$$

Issues with this result

We identify a few problems with the result and its proof:

- When we use the full codebook ($N = 2^n$ distinct known plaintexts, and $B = 0$), we can use any approximation as long as $\rho \neq 0$, as the right and wrong key variances always scale by the same amount related to ρ.
- The wrong-key correlation distribution is computed using the original key recovery map f, which should be irrelevant in the wrong-key setting.
- It is implicitly assumed that $g^{\perp}$ is always exactly balanced when sampled with real cipher data, when in reality it should have a randomized correlation.

For these reasons we decided to revise the proof of the result.

2 Sketch 2 Proof

For the right key, under live cipher data and randomized key, we assume:

$$f(x) \sim_K \|f\|_2 \cdot \mathcal{N}(0, \mathrm{ELP}) \qquad g^{\perp}(x) \sim_K \|g^{\perp}\|_2 \cdot \mathcal{N}(0, 2^{-n})$$

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$$\mathrm{Var}_{\mathcal{D},K}\left(\widetilde{\mathrm{cor}}_R^g\right) = \mathrm{Var}_{\mathcal{D},K}\left(\frac{\langle f, g\rangle}{\|f\|_2^2}\widetilde{\mathrm{cor}}_R^f\right) + \mathbf{Var}_{\boldsymbol{K}}\left(\mathbf{E}_{\boldsymbol{\mathcal{D}}}\left[\widetilde{\mathrm{cor}}_{\boldsymbol{R}}^{\boldsymbol{g}^{\perp}}\right]\right)$$

$$+ \mathrm{E}_K\left[\mathrm{Var}_{\mathcal{D}}\left(\widetilde{\mathrm{cor}}_R^{g^{\perp}}\right)\right] - 2\mathrm{Cov}_{\mathcal{D},K}\left(\frac{\langle f, g\rangle}{\|f\|_2^2}\widetilde{\mathrm{cor}}_R^f, \widetilde{\mathrm{cor}}_R^{g^{\perp}}\right)$$

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$$+ \mathrm{E}_K\left[\mathrm{Var}_{\mathcal{D}}\left(\widetilde{\mathrm{cor}}_R^{g^{\perp}}\right)\right] - \cancel{2\mathrm{Cov}_{\mathcal{D},K}\left(\frac{\langle f, g\rangle}{\|f\|_2^2}\widetilde{\mathrm{cor}}_R^{f}, \widetilde{\mathrm{cor}}_R^{g^{\perp}}\right)}$$

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For the right key, under live cipher data and randomized key, we assume:

$$f(x) \sim_K \|f\|_2 \cdot \mathcal{N}(0, \mathrm{ELP}) \qquad g^{\perp}(x) \sim_K \|g^{\perp}\|_2 \cdot \mathcal{N}(0, 2^{-n})$$

$$\begin{aligned}
\mathrm{Var}_{\mathcal{D},K}\left(\widetilde{\mathrm{cor}}_R^{g}\right) &= \mathrm{Var}_{\mathcal{D},K}\left(\frac{\langle f, g\rangle}{\|f\|_2^2}\widetilde{\mathrm{cor}}_R^{f}\right) + \mathbf{Var}_{\boldsymbol{K}}\left(\mathbf{E}_{\boldsymbol{\mathcal{D}}}\left[\widetilde{\mathrm{cor}}_{\boldsymbol{R}}^{\boldsymbol{g}^{\perp}}\right]\right) \\
&\quad + \mathrm{E}_K\left[\mathrm{Var}_{\mathcal{D}}\left(\widetilde{\mathrm{cor}}_R^{g^{\perp}}\right)\right] - \cancel{2\mathrm{Cov}_{\mathcal{D},K}\left(\frac{\langle f, g\rangle}{\|f\|_2^2}\widetilde{\mathrm{cor}}_R^{f}, \widetilde{\mathrm{cor}}_R^{g^{\perp}}\right)} \\
&\simeq \frac{\langle f, g\rangle^2}{\|f\|_2^2}\left(\frac{B}{N} + \mathrm{ELP}\right) + \|\boldsymbol{g}^{\perp}\|_2^2 \cdot \mathbf{2}^{-\boldsymbol{n}} + \|g^{\perp}\|_2^2\frac{B}{N}
\end{aligned}$$

2 Sketch 2 Proof

For the right key, under live cipher data and randomized key, we assume:

$$f(x) \sim_K \|f\|_2 \cdot \mathcal{N}(0, \mathrm{ELP}) \qquad g^{\perp}(x) \sim_K \|g^{\perp}\|_2 \cdot \mathcal{N}(0, 2^{-n})$$

$$\begin{aligned}
\mathrm{Var}_{\mathcal{D},K}\left(\widetilde{\mathrm{cor}}_R^{g}\right) &= \mathrm{Var}_{\mathcal{D},K}\left(\frac{\langle f, g\rangle}{\|f\|_2^2}\widetilde{\mathrm{cor}}_R^{f}\right) + \mathbf{Var}_{\boldsymbol{K}}\left(\mathbf{E}_{\boldsymbol{\mathcal{D}}}\left[\widetilde{\mathrm{cor}}_{\boldsymbol{R}}^{\boldsymbol{g}^{\perp}}\right]\right) \\
&\quad + \mathrm{E}_K\left[\mathrm{Var}_{\mathcal{D}}\left(\widetilde{\mathrm{cor}}_R^{g^{\perp}}\right)\right] - \cancel{2\mathrm{Cov}_{\mathcal{D},K}\left(\frac{\langle f, g\rangle}{\|f\|_2^2}\widetilde{\mathrm{cor}}_R^{f}, \widetilde{\mathrm{cor}}_R^{g^{\perp}}\right)} \\
&\simeq \frac{\langle f, g\rangle^2}{\|f\|_2^2}\left(\frac{B}{N} + \mathrm{ELP}\right) + \|\boldsymbol{g}^{\perp}\|_2^2 \cdot \mathbf{2}^{-\boldsymbol{n}} + \|g^{\perp}\|_2^2\frac{B}{N} \\
&= \|g\|_2^2\left(\frac{B}{N} + \rho^2 \cdot \mathrm{ELP} + (\mathbf{1} - \boldsymbol{\rho}^2)\mathbf{2}^{-\boldsymbol{n}}\right)
\end{aligned}$$

2 Sketch 2 Proof *(cont.)*

For the wrong key, we ignore the decomposition of g and just assume:

$$g \sim_K \|g\|_2 \cdot \mathcal{N}(0, 2^{-n})$$

2 Sketch 2 Proof *(cont.)*

For the wrong key, we ignore the decomposition of g and just assume:

$$g \sim_{\mathcal{K}} \|g\|_2 \cdot \mathcal{N}(0, 2^{-n})$$

$$\mathrm{Var}_{\mathcal{D},\mathcal{K}}\left(\widetilde{\mathrm{cor}}_W^g\right) = \mathrm{Var}_{\mathcal{K}}\left(\mathrm{E}_{\mathcal{D}}\left[\widetilde{\mathrm{cor}}_W^g\right]\right) + \mathrm{E}_{\mathcal{K}}\left[\mathrm{Var}_{\mathcal{D}}\left(\widetilde{\mathrm{cor}}_W^g\right)\right]$$

2 Sketch 2 Proof *(cont.)*

For the wrong key, we ignore the decomposition of g and just assume:

$$g \sim_K \|g\|_2 \cdot \mathcal{N}(0, 2^{-n})$$

$$\begin{aligned}\mathrm{Var}_{\mathcal{D},K}\left(\widetilde{\mathrm{cor}}_W^g\right) &= \mathrm{Var}_K\left(\mathrm{E}_{\mathcal{D}}\left[\widetilde{\mathrm{cor}}_W^g\right]\right) + \mathrm{E}_K\left[\mathrm{Var}_{\mathcal{D}}\left(\widetilde{\mathrm{cor}}_W^g\right)\right] \\ &\simeq \|g\|_2^2 \left(\frac{B}{N} + 2^{-n}\right)\end{aligned}$$

We note that, save for the normalization factor $\|g\|_2$, the wrong key distribution is the same as for the attack before substituting f, which is different from the previous version in the result. In addition, the right key variance is higher than expected due to the noise $((1-\rho^2)2^{-n})$ introduced by $g^{\perp}$.

We have confirmed through experiments that this version is more accurate.